1M1B · AI FOR SUSTAINABILITY

SDG 12 SDG 13

AI Household Sustainability Advisor

EcoGranite

Helping households audit resources, plan toward net-zero, and learn sustainable habits through conversational AI.

SUBMITTED BY

Dhanish Ladwani

In collaboration with IBM SkillsBuild & AICTE

AT A GLANCE

Solution Overview

EcoGranite is a web platform that helps households measure their environmental footprint and act on it.

It combines a carbon assessment, a net-zero planner, a waste-sorting game, and an AI advisor into one connected experience — turning sustainability from an abstract idea into daily, measurable steps.

4 interactive sustainability tools

1 IBM Granite AI advisor at the core

2 UN Sustainable Development Goals

UNITED NATIONS GOALS

SDG Alignment

12 Responsible Consumption & Production

Guides households to audit daily consumption, cut waste, and adopt efficient resource use through measurable feedback.

Targets 12.5 · 12.8 — waste reduction & sustainability awareness

13 Climate Action

Quantifies household carbon footprints and sets net-zero goals, translating climate awareness into concrete domestic action.

Target 13.3 — climate education & capacity building

THE CHALLENGE

Problem Statement

> How might we use AI to guide domestic resource auditing, goal planning, and gamified sorting education so households and local communities become more sustainable in their daily consumption and waste behaviors?

~66%

of global emissions are linked to household consumption choices.

Daily

decisions on energy, waste, and water lack clear, personal feedback.

Gap

between climate awareness and practical, measurable home action.

EMPATHIZE

User Insights

”I don’t actually know my household’s carbon footprint.”	→ The Sustainability Index makes the footprint visible and scored.
”Recycling and sorting rules are confusing.”	→ The Eco-Sorter game teaches correct sorting through play.
”Generic climate advice doesn’t fit my home.”	→ The AI Advisor gives personalized, context-aware guidance.
”I have no way to set or track real goals.”	→ The Net-Zero Planner sets targets and tracks progress over time.

WHO IT SERVES

Target Users

Households

Families and individuals who want to understand and lower their energy, water, and waste footprint at home with simple guidance.

Local Communities

Neighborhoods and groups building shared sustainable habits around recycling, conservation, and collective net-zero goals.

Students & Learners

Young people learning waste sorting and climate concepts through interactive, gamified, and conversational experiences.

Solution type: Conversational AI Assistant combined with a Decision-Support System for everyday sustainability.

OUR PROCESS

Design Thinking

01

Empathize

Studied how households struggle to track waste and energy habits.

02

Define

Framed the need for personal auditing and clear goal feedback.

03

Ideate

Brainstormed an AI advisor, index, planner, and sorting game.

04

Prototype

Built the SPA with Granite AI integration and live features.

05

Test & Refine

Iterated on accuracy, usability, and responsible AI safeguards.

AI Solution

POWERED BY

IBM watsonx.ai

Granite-3-8b-instruct

A foundation model accessed securely through IBM Cloud IAM authentication and a server-side proxy.

Conversational guidance

Natural-language answers to household sustainability questions.

Decision support

Personalized tips based on the user's footprint and goals.

Contextual prompting

Prompt engineering frames Granite as a focused eco-advisor.

UNDER THE HOOD

Technology Stack

Frontend

HTML & CSS

JavaScript

Glassmorphic UI

Single-page app

Backend

Python

Flask server

REST /api/chat

Secure proxy

AI & Cloud

IBM watsonx.ai

Granite-3-8b

IBM Cloud IAM

Token auth

Data

Browser storage

localStorage

No server DB

Privacy-first

Design choice: a lightweight client with all credentials kept server-side keeps the app fast, portable, and secure.

HOW IT WORKS

System Architecture

FRONTEND

Glassmorphic SPA

HTML, CSS & JavaScript interface with the dashboard, planner, sorting game, and chat. State stored locally in the browser.

BACKEND

Flask Proxy

A Python server exposes /api/chat, requests an IBM IAM token, and securely relays prompts — keeping credentials off the client.

AI LAYER

IBM watsonx.ai

Granite-3-8b-instruct processes the prompt and returns sustainability guidance back through the proxy to the user.

User → SPA → Flask /api/chat → IAM Token → Granite → Response

WHAT IT DOES

Key Features

01

Sustainability Index

Carbon assessment scoring a household's overall footprint.

02

Net-Zero Goal Planner

Set and track reduction targets on the road to net-zero.

03

Eco-Sorter Game

A gamified conveyor that teaches correct waste sorting.

04

AI Advisor Chat

Granite-powered conversational eco-guidance on demand.

05

Theme Customizer

Multi-accent personalization for an inclusive experience.

One connected home for sustainable habits

FEATURE DEEP-DIVE

Measure & Plan

01

Sustainability Index

Users enter energy, water, and waste habits. The app calculates a carbon footprint and turns it into a single, easy-to-read sustainability score that updates as habits change.

02

Net-Zero Goal Planner

Households set reduction targets and track progress over time. The planner breaks the path to net-zero into achievable milestones that keep motivation high.

FEATURE DEEP-DIVE

Learn & Advise

03

Eco-Sorter Game

A conveyor-belt game where players drop items into the right bins. Correct sorting builds intuition for recycling rules, making waste education hands-on and memorable.

04

AI Advisor Chat

Powered by IBM Granite, the advisor answers sustainability questions in natural language and suggests next steps tailored to the user's footprint and goals.

ETHICS BY DESIGN

Responsible AI

Transparency

Users know they are talking to an AI advisor, with clear context on how guidance is generated.

Privacy

Personal data stays in the browser via localStorage; API keys remain server-side, never exposed to the client.

Fairness

Guidance is inclusive and accessible, with customization that adapts to diverse users and preferences.

Safety

Prompts are scoped to sustainability topics, keeping responses relevant, constructive, and reliable.

THE OUTCOME

Expected Impact

Environmental

Lower household carbon footprints through measurable, sustained reductions in energy and waste.

Social

Greater climate literacy and sustainable habits across families, students, and communities.

Economic

Real household savings from reduced energy and water consumption over time.

> EcoGranite turns climate awareness into everyday action — making sustainable living measurable, guided, and achievable at home.

WHAT'S NEXT

Future Roadmap

NEAR TERM

Smarter Tracking

Historical trends, reminders, and richer charts so households can see progress week over week.

MID TERM

Community Mode

Shared goals, leaderboards, and neighborhood challenges to build collective momentum.

LONG TERM

Real Data Links

Integrations with utility data and smart meters for automatic, accurate footprint tracking.

> The goal: move from self-reported estimates to automatic, community-wide sustainability action.

THANK YOU

SDG 12 SDG 13

EcoGranite

A practical AI advisor guiding households toward responsible consumption and climate action — one decision at a time.

PRESENTED BY

Dhanish Ladwani

1M1B AI for Sustainability · IBM SkillsBuild & AICTE